

End-to-End Process · Idea-to-Market (I2M)

PCF 2.1 Executive Steering Committee · PCF 2.2 R&D / Innovation · PCF 2.3.1–2.3.3 Engineering, Marketing & Operations

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View 1 — HITL Design

Pattern Type Legend

Decision 80% H at M3	Knowledge 65% H	Document 40% H	Transaction 35% H	Exception 82% H
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I2M End-to-End Process — HITL Design · PCF-Derived Process Ownership

Executive Steering Committee PCF 2.1	R&D / Innovation PCF 2.2	Engineering / Design PCF 2.3.1	Marketing PCF 2.3.2	Operations PCF 2.3.3
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HITL Design Insight

HITL Design Insight — I2M

2 Decision gates (80% Human at M3) sit with the Executive Steering Committee — portfolio governance and lifecycle management carry capital-allocation authority that stays irreducibly human, even though a parts/accessories variant of the portfolio decision routes to Product Management at lower stakes (documented as an exception rather than a parallel path). **4 Knowledge steps** (65% Human at M3) span discovery research, concept generation, requirements definition, and design/prototyping — the largest cluster, and where AI augmentation of research synthesis and generative design tooling offers the clearest opportunity. Design/prototyping (2.3.1) also carries an **inline Stage-Gate Committee approval** — a real decision checkpoint with no ongoing work of its own, so it's marked on the step rather than given its own lane, and excluded from the aggregate H% math. **1 Document step** (40% Human at M3) covers market-test synthesis and business-case finalization. **1 Transaction step** (35% Human at M3) — production-readiness preparation — shows a smaller M1→M3 gap than a typical transactional step because mature PLM/MES systems already automate much of this work independent of any AI involvement. I2M aggregate at M3: **61.9% Human / 38.1% AI** — reflecting a process where executive judgment and engineering/creative synthesis remain central even as AI absorbs more of the research and production-prep surface.

HITL Impact Analysis

H% = Human involvement at each maturity level · M3 = HITL Design Intent · ★ = Design target

Process Step (L3)	PCF	Pattern	M1 H%	M3 H% ★	Δ M1→M3	Opportunity
I2M End-to-End Process 2.1–2.3.3	—	Mixed	72.1%	61.9%	-10.2%	Significant
Executive Steering Committee (PCF 2.1)						
Manage Product & Service Portfolio	2.1.1	Decision	88%	80%	-8%	Retain Human
Manage Product & Service Life Cycle	2.1.2	Decision	85%	80%	-5%	Retain Human
R&D / Innovation (PCF 2.2)						
Perform Discovery Research	2.2.1	Knowledge	80%	65%	-15%	High

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Generate New Product/Service Concepts	2.2.2	Knowledge	75%	65%	-10%	Moderate
Define Product/Service Development Requirements	2.2.3	Knowledge	73%	65%	-8%	Low
Engineering / Design (PCF 2.3.1)						
Design & Prototype Products/Services (+ inline Stage-Gate Committee approval — excluded from H% math)	2.3.1	Knowledge	78%	65%	-13%	Moderate
Marketing (PCF 2.3.2)						
Test Market for New/Revised Products & Services	2.3.2	Document	55%	40%	-15%	High
Operations (PCF 2.3.3)						
Prepare for Production/Service Delivery	2.3.3	Transaction	43%	35%	-8%	Low

AI Capability Map — I2M

PCF 2.1 Executive Steering Committee		PCF 2.2 R&D / Innovation	
2.1.1 Portfolio scenario & market-gap analysis	DECISION	2.2.1 Technology & trend discovery synthesis	KNOWLEDGE
2.1.2 Lifecycle performance & retirement modeling	DECISION	2.2.2 Concept generation & ideation support	KNOWLEDGE
M1: 87% M3: 80% M5: 68%		2.2.3 Requirements drafting & feasibility synthesis	KNOWLEDGE
		M1: 76% M3: 65% M5: 52%	
PCF 2.3.1 Engineering / Design		PCF 2.3.2 Marketing	
2.3.1 Generative design & prototyping support	KNOWLEDGE	2.3.2 Market-study & customer-interview synthesis	DOCUMENT
2.3.1 Design-spec drafting & documentation	KNOWLEDGE	2.3.2 Business case drafting support	DOCUMENT
M1: 78% M3: 65% M5: 50%		M1: 55% M3: 40% M5: 28%	
PCF 2.3.3 Operations			
2.3.3 Production-readiness checklisting (PLM/MES)	TRANSACTION		
2.3.3 Engineering-change routing automation	TRANSACTION		
M1: 43% M3: 35% M5: 24%			